

Navigating the Pauson–Khand Reaction in Total Syntheses of Complex Natural Products

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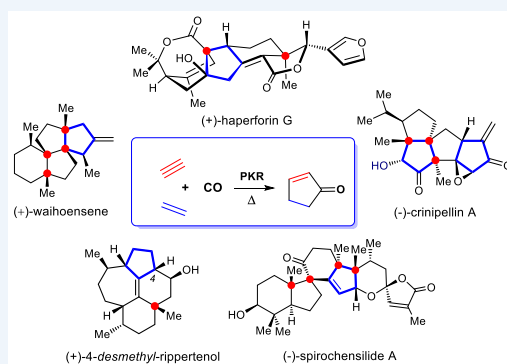
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CONSPECTUS: “Total synthesis endeavors provide wonderful opportunities to discover and invent new synthetic reactions as a means to advance organic synthesis in general. Such discoveries and inventions can occur when the practitioner faces intransigent problems that cannot be solved by known methods and/or when method improvements are desired in terms of elegance, efficiency, cost-effectiveness, practicality, or environmental friendliness” (K. C. Nicolaou et al. from their review in *CCS Chem.* 2019, 1, 3–37). To date tens of thousands of bioactive compounds have been isolated from plants, microbes, marine invertebrates, and other sources. These chemical structures have been studied by chemists who scanned the breadth of natural diversity toward drug discovery efforts. Drug-likeness of natural products often possesses common features including molecular complexity, protein-binding ability, structural rigidity, and three-dimensionality. Considering certain biologically important natural products are scarce from natural supply, total synthesis may provide an alternative solution to generating these compounds and their derivatives for the purpose of probing their biological functions. Natural products bearing quaternary carbon stereocenters represent a group of biologically important natural entities that are lead compounds in the development of pharmacological agents and biological probes. However, the stereocontrolled introduction of quaternary carbons, with vicinal patterns that substantially expand the complexity of molecular architectures and chemical space in particular, presents distinct challenges because of the high steric repulsion between substituents. Though remarkable advance has been seen for quaternary carbon stereocenter generation, the process remains a daunting challenge given that the formation of highly congested stereocenters increases the difficulty in achieving orbital overlap.

In the past two decades, our group has initiated a program to develop synthetic strategies and methods with the aim of advancing the frontiers of the total syntheses of biologically important complex natural products bearing all-carbon quaternary stereogenic centers. Typical endeavors have involved the use of a Pauson–Khand (PK) reaction as a key step in constructing core structures with all-carbon quaternary stereogenic center(s), with the aid of well-orchestrated thiourea–Co- and thiourea–Pd-catalyzed PK reactions. These methodological advances have enabled us to achieve total syntheses of a series of topologically complex natural products with diverse structural features. These methods will enable the assembly of molecules with improved biological functions and provide tool compounds for elucidation of mechanism of action or identification of potential cellular targets.



KEY REFERENCES

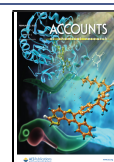
- Qu, Y.-Z.; Wang, Z.-Y.; Zhang, Z. C.; Zhang, W.-D.; Huang, J.; Yang, Z. Asymmetric Total Synthesis of (+)-Waihoensene. *J. Am. Chem. Soc.* 2020, 142, 6511–6515.¹ This work presents an asymmetric total synthesis of (+)-waihoensene by using an intramolecular Pauson–Khand reaction and a radical-initiated intramolecular hydrogen atom transfer (HAT) reaction as key steps.
- Liang, X.-T.; Chen, J.-H.; Yang, Z. Asymmetric Total Synthesis of (–)-Spirochensilide A. *J. Am. Chem. Soc.* 2020, 142, 8116–8121.² The work describes an asymmetric total synthesis of (–)-spirochensilide A, and the key steps are a semipinacol rearrangement reaction, a

tungsten-mediated cyclopropene-based Pauson–Khand reaction, and a furan-based oxidative cyclization reaction.

- Zhang, Z.-C.; Zhao, D.-D.; Zhang, Z.-C.; Tan, X.-Y.; Gong, J.-X.; Fu, J.-K.; Yang, Z. Synthesis of 4-Desmethyl-rippertanol and 7-Epi-rippertanol via Photo-induced

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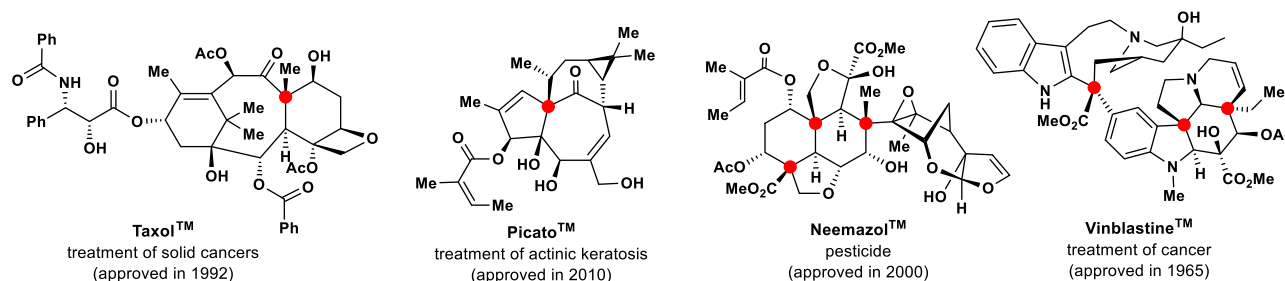


Figure 1. Selected biologically important natural products bearing all-carbon quaternary stereogenic centers.

Cyclization of Dienones. *CCS Chem.* **2020**, *2*, 2074–2083.³ This work illustrates concise syntheses of 4-desmethyl-rippertenol and 7-epi-rippertenol based on an intramolecular Pauson–Khand reaction and a novel photoinduced intramolecular cyclization of α -cyclopropyl dienone as key steps.

- Zhang, W.; Zhang, Z.-Y.; Tang, J.-C.; Che, J.-T.; Zhang, H.-Y.; Chen, J.-H.; Yang, Z. Total Synthesis of (+)-Haperforin G. *J. Am. Chem. Soc.* **2020**, *142*, 19487–19492.⁴ This work discloses an asymmetric total synthesis of (+)-haperforin G and the discovery of a late-stage cross-coupling reaction of unstabilized C(sp³)-radical with enone via a light-initiated photocatalysis.

■ INTRODUCTION

Quaternary carbon stereocenters occur frequently in natural products, drugs, and bioactive molecules (Figure 1). For example, 12% of the top 200 prescription drugs sold in the U.S.A. in 2011 contain quaternary carbon stereocenters.⁵ The tetrahedral nature of all-carbon quaternary stereogenic center(s),⁶ particularly in natural products that coevolved with protein-binding partners, can confer on natural products well-defined topological complexity and conformational constraints. These decrease the conformational entropy penalty on binding to a protein target⁷ and are exemplified by many highly impactful natural products, such as Taxol, Picato, Neemazol, and Vinblastine (Figure 1). However, the enantioselective construction of quaternary carbon stereocenters represents a long-standing challenge, and intensive efforts have been made to develop synthetic strategies and methods for creating such stereogenic centers.⁸

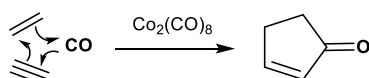
The Pauson–Khand (PK) reaction⁹ is a [2 + 2 + 1] cycloaddition among an alkyne, an alkene, and carbon monoxide to yield cyclopentenones (Scheme 1). Despite its synthetic

importance, the original PK reaction needed improvements to enhance its reaction rates and yields and development of environmentally friendly versions to circumvent the use of stoichiometric amounts of the toxic and moisture sensitive reagent Co₂(CO)₈. Improvements in terms of reactivity, selectivity, and catalysis were gradually achieved by using additives and other transition-metal-based complexes to accelerate and catalyze this type of reaction.¹⁰

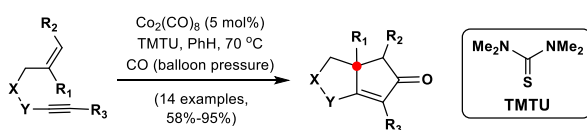
In recent decades, our group has focused on continuing the search for improved methods for increasing the PK reaction efficiency. We found that a new complex, that is, Co₂(CO)₈/tetramethyl thiourea (TMTU),¹¹ catalyzed PK reactions under a balloon pressure of CO at 70 °C (Figure 2A-1). This enabled carbonylative cyclization of a broad range of enynes to the corresponding cyclopentenones and was used in the total syntheses of complex natural products by us¹² and other groups.¹³ Two important features account for the attractiveness of this efficient protocol. (1) The reaction proceeds with a catalytic amount of Co₂(CO)₈ (5 mol %) under balloon pressure of CO to give the products in good to excellent yields. (2) When R₁ is an alkyl group, this reaction can afford a product bearing an all-carbon quaternary chiral center at a bridge head position, and this provides an alternative pathway for constructing this type of synthetically intriguing stereogenic center. We explored more robust reagents and found that TMTU/PdCl₂¹⁴ (Figure 2A-2) and TMTU/CoBr₂¹⁵ were less toxic, were air and moisture stable, and smoothly catalyzed certain types of PK reactions.

In our previous Account,¹⁶ we described our efforts to develop total syntheses of a group of structurally intriguing nortriterpenoids (Figure 3) 1–4 bearing all-carbon quaternary chiral center(s). The cyclopentenone motifs in 1–3 were constructed via PK reactions as key steps. In this Account, we trace the evolution of our use of PK reactions as key steps in building the cyclopentenone¹⁷ motifs in natural products 5–9 (Figure 3) and highlight how strategic cyclopentenone constructions can streamline the total syntheses of complex natural products and contribute to both the science and art of total synthesis campaigns.

Scheme 1. General Pauson–Khand Reaction



A-1. Tetramethyl Thiourea (TMTU)/Co₂(CO)₈-catalyzed Pauson–Khand reaction



A-2. Tetramethyl Thiourea/PdCl₂-catalyzed Pauson–Khand reaction

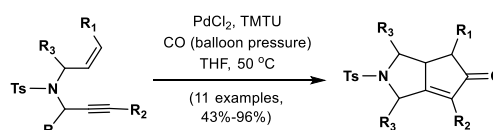


Figure 2. Two enabling synthetic methodologies for improving PK reaction efficiency.

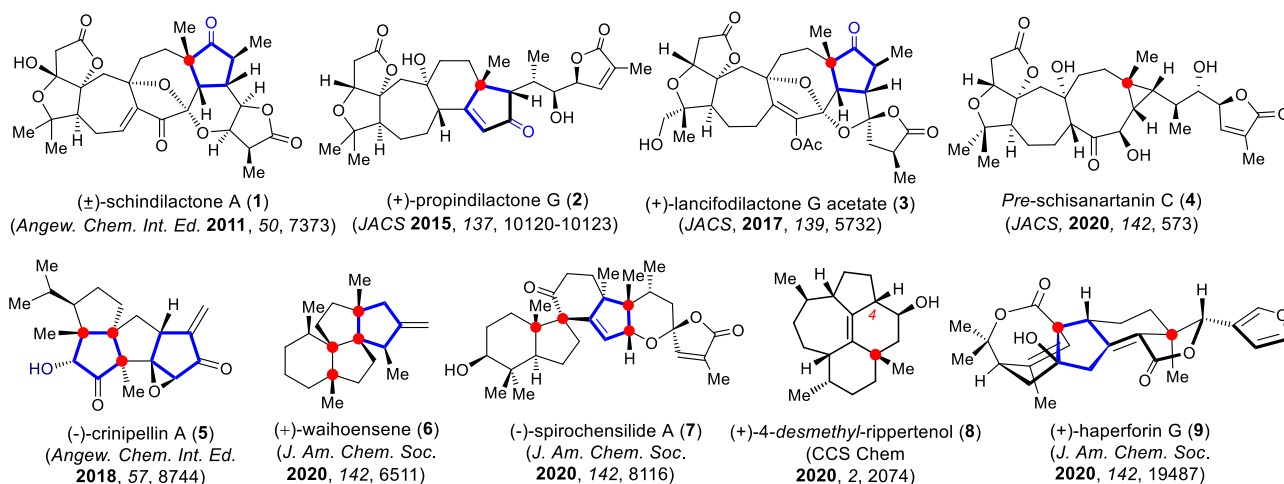


Figure 3. Selected natural products synthesized by our group.

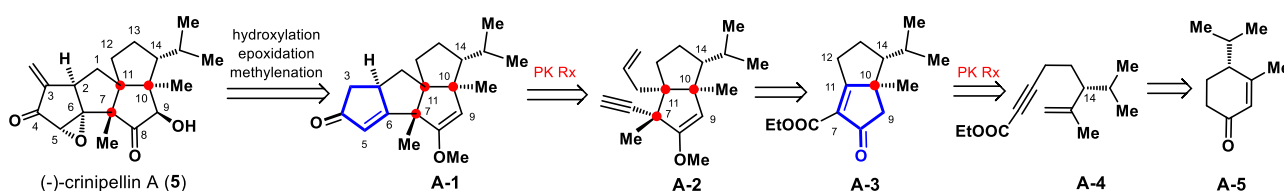
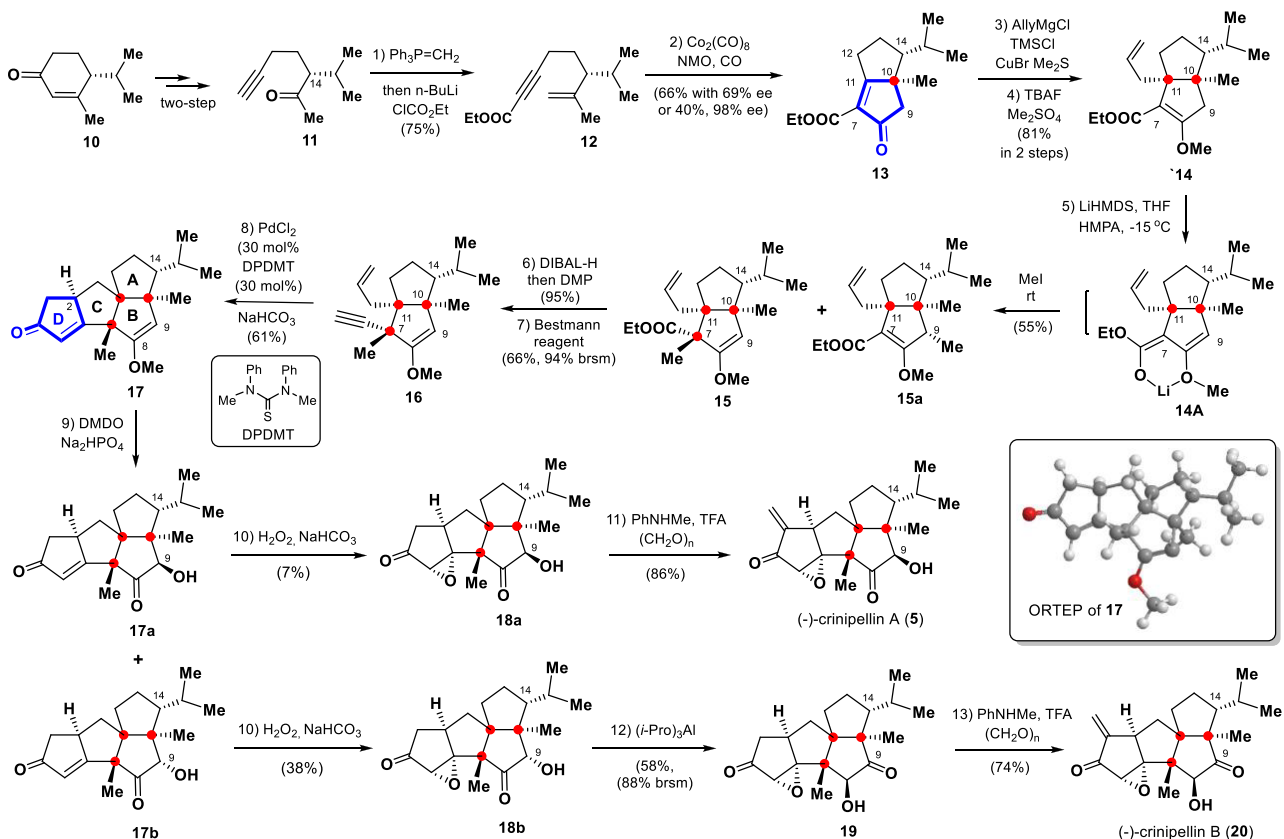


Figure 4. Retrosynthetic analysis of crinipellin A (5).

Scheme 2. Total Synthesis of Crinipellin A (5)

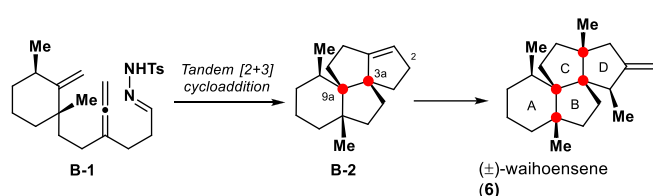


■ TOTAL SYNTHESIS OF CRINIPELLIN A

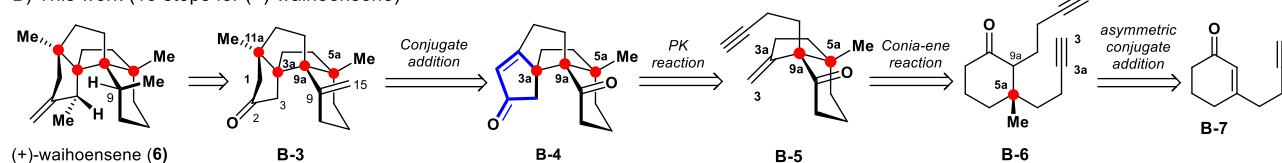
Crinipellin A (5, Figure 3) is one of several related diterpenoids isolated from the fungus *Crinipellis stipitaria* (Agaricales) by Steglich and co-workers in 1979.^{18,19} The structure of 5 includes

a signature α -*exo*-methylene ketone motif in conjunction with a rare and contiguous sequence of three quaternary carbon atoms within a tetraquinane core. Biologically, 5 has antibiotic activity,²⁰ and its C9 acetate can completely inhibit the

A) Total synthesis of (±)-waihoensene by Lee and co-workers in 2017



B) This work (15 steps for (+)-waihoensene)

**Figure 5.** Retrosynthetic analysis of waihoensene (6).

biosyntheses of DNA, RNA, and proteins in Ehrlich carcinoma cells at a concentration of 5 mg/mL. The α -*exo*-methylene ketone motif makes **5** a potential irreversible probe²¹ in the field of chemical biology.

Retrosynthetic analysis of our total synthesis²² suggested that **5** could be made from **A-1** via a hydroxylation, epoxidation, and methylenation sequence (Figure 4). The cyclopentenone motifs in **A-1** and **A-3** could be formed via intramolecular PK reactions from the enynes **A-2** and **A-4**, respectively. The key intermediate enyne **A-2** could be constructed from **A-3** by diastereoselective installation of two vicinal quaternary stereogenic centers, a process that would probably involve substrate conformational control. We planned to obtain **A-4** bearing the C14 stereogenic center from **A-5** by using Trost's protocol.²³

As shown in Scheme 2, a Wittig reaction of ketone **11** and then carbonylation afforded enyne **12** in 75% yield. The PK reaction was achieved by treatment of **12** (bearing an electron-deficient alkyne²⁴) with a stoichiometric amount of $\text{Co}_2(\text{CO})_8$ at room temperature in dichloroethane (DCE) for 1 h. The resultant Co-complex was gradually warmed up to 75 °C in the presence of 4-methylmorpholine *N*-oxide (NMO) for 36 h to give ketoester **13** in 66% yield with 69% ee (40% yield with 98% ee after crystallization).

The decrease in the ee value was presumably the result of epimerization of the C14 stereochemistry during the Wittig reaction. Diastereoselective conjugate addition of **13** with an organocopper reagent (derived from allyl magnesium chloride and $\text{CuBr}\cdot\text{Me}_2\text{S}$) proceeded at −78 °C. Reaction of the resultant enolate with Me_2SO_4 in the presence of tetrabutylammonium fluoride (TBAF) afforded the methyl vinyl ether **14** in 81% yield as a single isomer. This set the stage for installation of the critical third vicinal quaternary chiral center at C7.

Compound **14** was treated with lithium bis(trimethylsilyl)-amide (LiHMDS) in the presence of hexamethylphosphoramide (HMPA) at −15 °C in tetrahydrofuran (THF), and the resultant enolate was reacted with MeI at room temperature to afford **15** in 55% yield, together with its regioisomer **15a** in 18% yield. Presumably **15a** was formed because **14A** can exist in chelated and unchelated forms at room temperature and the latter is more reactive at the α -position. When this reaction was performed at −78 °C, **15a** was obtained in 67% yield as the sole product, as a result of the Curtin–Hammett effect,²⁵ because the α -position is more hindered than the γ -position in complex **14A**.

Enyne **16** was prepared via a three-step sequence (Scheme 2). Reduction of the ethyl ester group in **15** with DIBAL-H and oxidation of the resultant alcohol (Dess–Martin reagent)

afforded an intermediate aldehyde, which was transformed to **16** (Bestmann reagent) in 66% overall yield. We next used the PK reaction to forge tetracycle **17**. We found that a novel complex, that is, PdCl_2 /diphenyldimethyl thiourea (DPDMT), catalyzed the PK reaction to afford the desired product **17** in 61% yield, together with its C2-isomer in 16% yield. Tetracycle **17** was amenable to single-crystal X-ray structural analysis.

With **17** in hand, we investigated its elaboration to achieve the total syntheses of targets **5** and **20**. Regioselective epoxidation of **17** with dimethyldioxirane (DMDO) and subsequent vinyl ether epoxide hydrolysis afforded **17a** and **17b**, which underwent diastereoselective epoxidation to give epoxides **18a** and **18b** in 7% and 38% yields, respectively. Treatment of **18a** with *N*-methylanilinium trifluoroacetate and paraformaldehyde in THF at 70 °C furnished (−)-crinipellin A [(−)-**5**] in 86% yield. The total synthesis of (−)-crinipellin B [(−)-**20**] was achieved via $\text{Al}(\text{iPrO})_3$ -mediated hydroxyl-ketone isomerization of **18b** and methylenation of the resultant **19** under identical conditions to those used to convert **18a** to (−)-**5** to give (−)-**20** in 74% yield.²⁶ The ¹H NMR and ¹³C NMR spectra and optical rotation data for synthetic (−)-crinipellin A [(−)-**5**] and (−)-crinipellin B [(−)-**20**] were in good agreement with those reported in the literature.^{18,27}

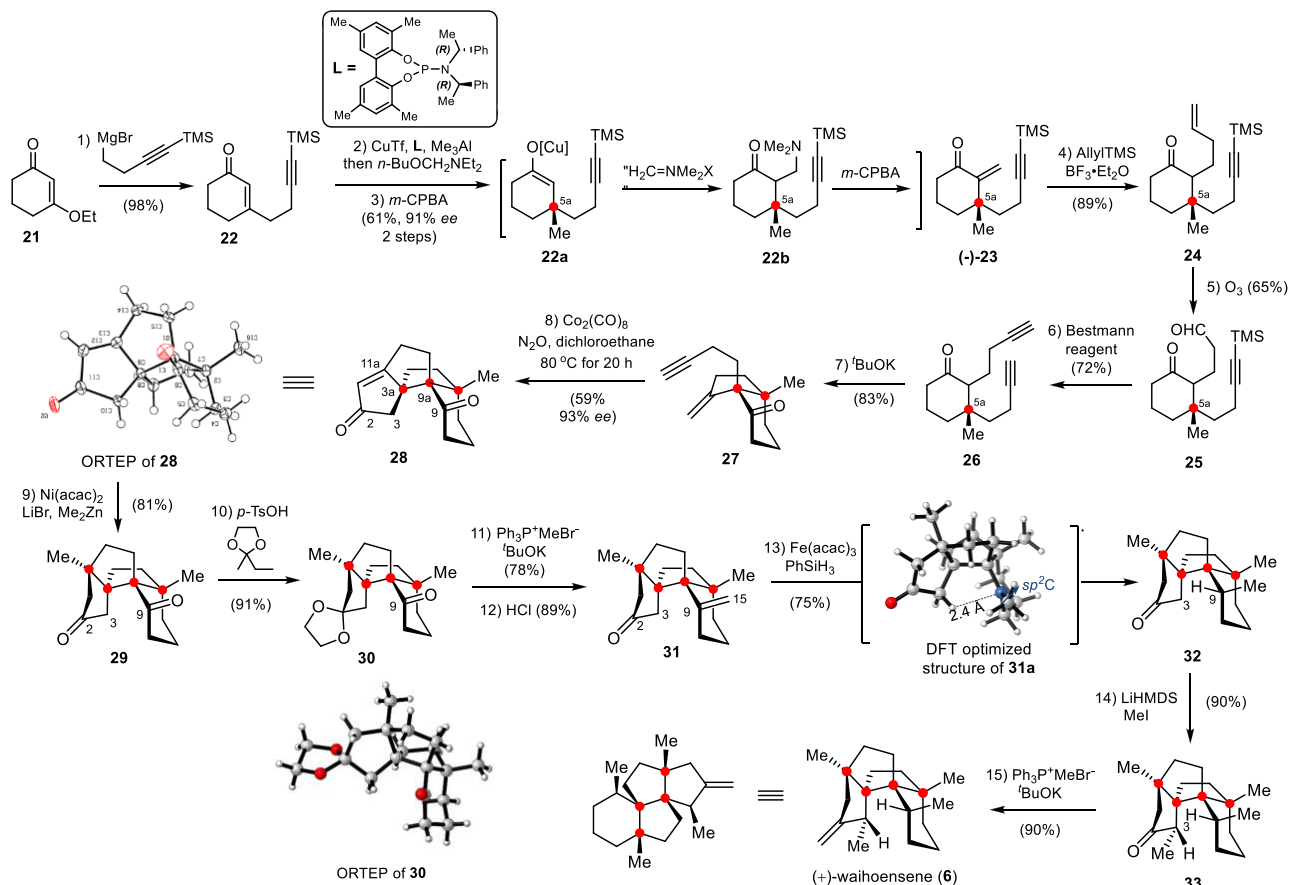
■ TOTAL SYNTHESIS OF (+)-WAIHOENSENE

(+)-Waihoensene **6**²⁸ was isolated in 1997 by Weavers and co-workers from the New Zealand podocarp, *Podocarpus totara* var. *waihoensis*. Molecule **6** is characterized by a congested and *cis*-fused tetracyclic core featuring six contiguous stereogenic centers, four of which are quaternary centers (C3a, C5a, C9a, and C11a) and are contiguously situated at the compact and bridged-head region of the molecule. Given its structural complexity, there have been many efforts to derive efficient synthesis of **6**, culminating in an impressive 18-step synthesis of (±)-waihoensene by the laboratory of Lee in 2017.²⁹ The synthesis features a tandem [2 + 3] cycloaddition to construct the BCD tricyclic ring with two contiguous quaternary stereogenic centers²⁹ (Figure 5A). Recently, we reported the asymmetric total synthesis of **6**,¹ which features an intramolecular hydrogen atom transfer (HAT) reaction for stereoselective installation and formation of the C9 stereogenic center (Figure 5B).

Figure 5B shows, in retrosynthetic format, the evolution of the synthetic strategy leading to (+)-waihoensene (**6**). The desired C9 stereogenic center in **6** was formed from **B-3** via an appropriate hydrogenation. According to Lee's total synthesis of

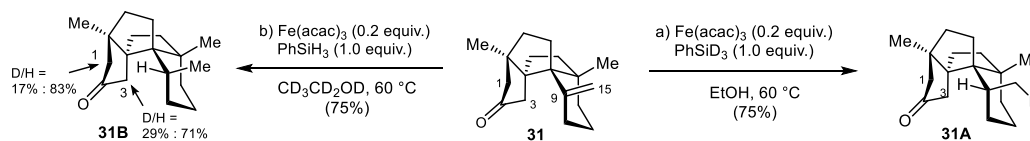
Scheme 3. Total Synthesis of (–)-Waihoensene (6) and Deuterium Labeling Studies and Control Experiments

A) Total synthesis of (–)-waihoensene (6)

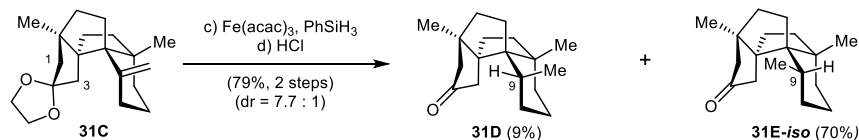


B) Deuterium labeling studies and control experiments

a. Deuterium labeling studies



b. Control experiment



waihoensene, the requisite C11a quaternary stereogenic center in intermediate **B-3** can be built up from **B-4** via a cuprate-mediated conjugate addition reaction in a highly diastereoselective manner (see the DFT optimized structure **B-4'**, and nucleophile will approach the enone motif from its bottom face). The cyclopentenone motif in **B-4** was expected to be derived from enyne **B-5** via a PK reaction because of the well-defined substrate conformation (see DFT optimized structure **B-5'**). The vicinal C9a and C5a quaternary chiral centers in the angular triquinane **B-5** could be constructed diastereoselectively from diyne **B-6** by a Conia-ene reaction, which in turn could be generated enantioselectively from enone **B-7** via a Cu-catalyzed conjugate addition.

As shown in Scheme 3, our synthesis commenced with development of an asymmetric approach to the enantioselective synthesis of diyne **26**. A Grignard reaction of enone **21** gave **22**, which underwent Cu-catalyzed asymmetric conjugate addition (ACA) by reaction with Me_3Al in the presence of the SimplePhos ligand **L**³⁰ to afford enolate **22a**, which without purification was reacted with the in situ generated electrophilic iminium ion followed by oxidation with *meta*-chloroperoxybenzoic acid (*m*CPBA) to give enone (–)-**23** in 61% yield with 91% ee. Allylation of enone **23** via a Sakurai reaction, and then ozonolysis furnished aldehyde **25** in 58% yield over two steps. The Ohira–Bestmann reaction of **25** afforded diyne **26** in 72% yield, with concomitant trimethylsilyl (TMS) desilylation.

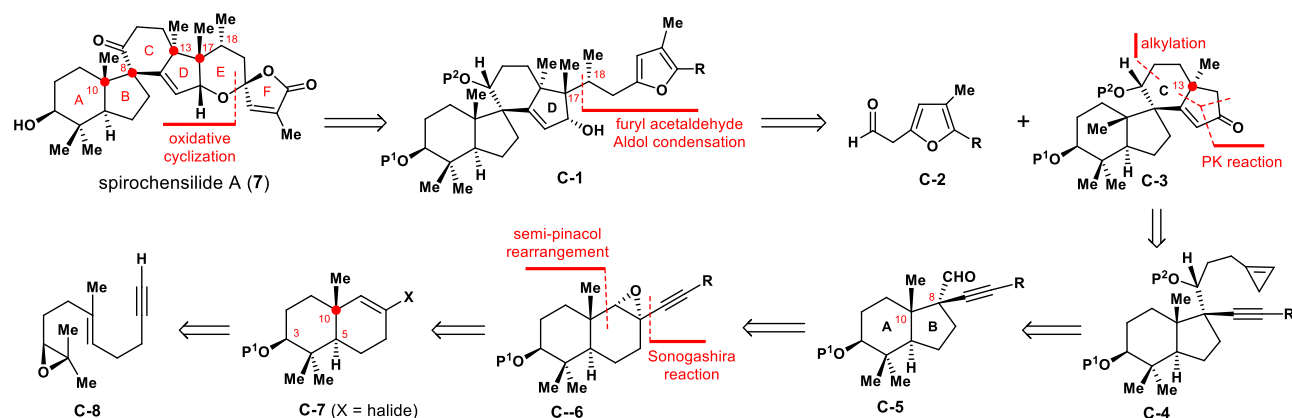
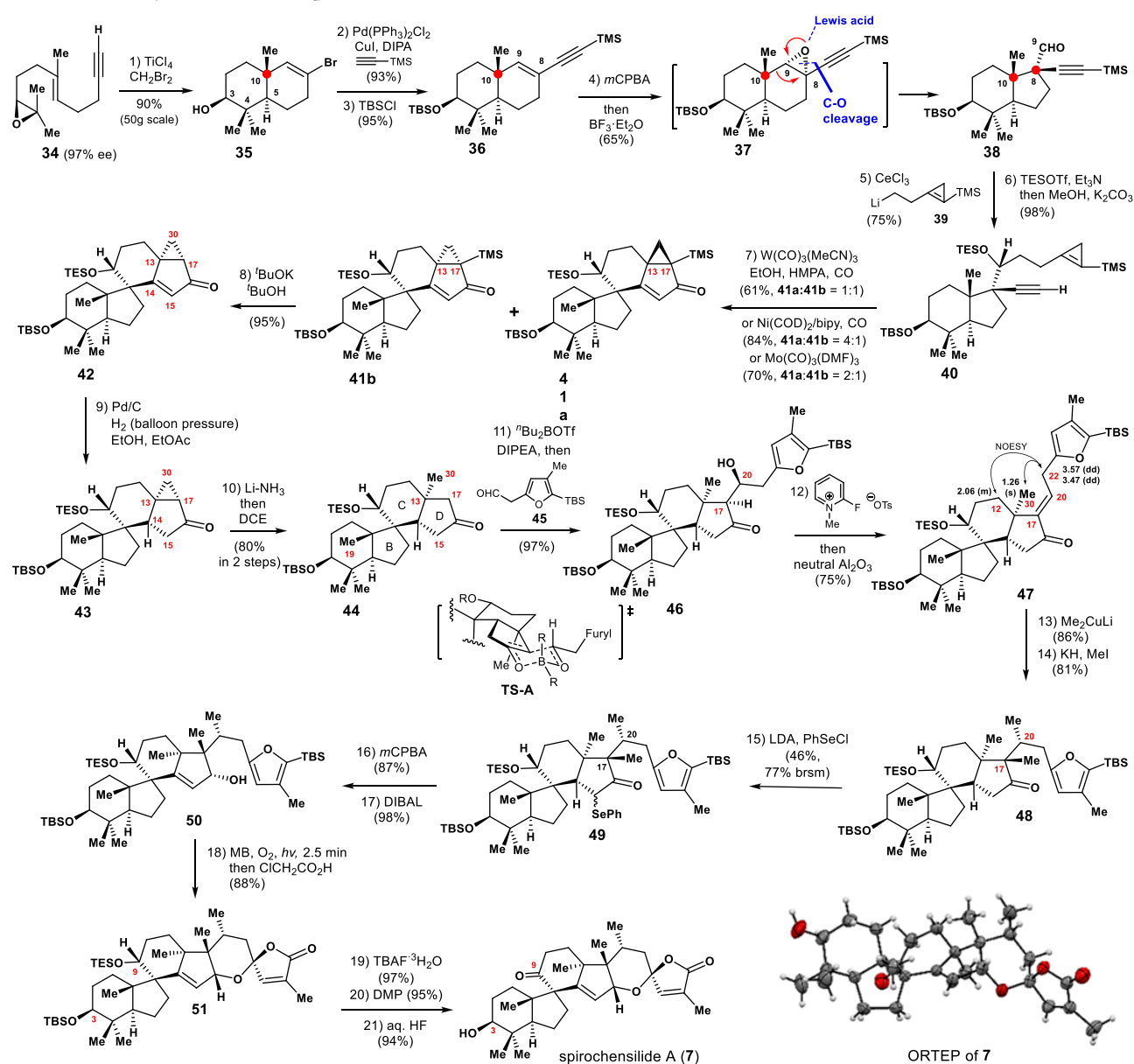


Figure 6. Retrosynthetic analysis of (-)-spirochensilide A (7).

Scheme 4. Total Synthesis of (-)-Spirochensilide A (7)



83% yield as a single diastereomer. A PK reaction of **27** was accomplished by treatment with $\text{Co}_2(\text{CO})_8$ in the presence of N_2O ³¹ to forge tetracycle **28** in 59% yield with 93% ee. The structure of **28** was confirmed by X-ray crystallographic analysis. Finally, **28** underwent stereoselective Ni-catalyzed methylation³² to deliver diketone **29** in 81% yield.

We next worked on regio- and diastereoselective methylation at the C3 and C9 positions. An acid-catalyzed ketal exchange reaction of **29** with 2-ethyl-2-methyl-1,3-dioxolane gave monoketal **30**, which was converted to ketone **31** via a Wittig reaction and subsequent ketal hydrolysis. The structure of **30** was confirmed by X-ray crystallographic analysis. Diastereoselective hydrogenation of **31** was achieved via an intramolecular [1,4]-HAT reaction by treatment with $\text{Fe}(\text{acac})_3$ as a catalyst in the presence of PhSiH_3 .³³ The desired product **32** was obtained in 75% yield as a single diastereoisomer. We reasoned that the resultant C9 carbon radical in **31a**, which was generated from **31** by a radical-mediated reductive reaction, could abstract a proton via intramolecular HAT reaction from C3, due to the close proximity (2.4 Å) of C3 and C9 and their position next to the C2 carbonyl group.

To support our analysis, we performed deuterium labeling studies and a control experiment. Accordingly, treatment of **31** with a catalytic amount of $\text{Fe}(\text{acac})_3$ in the presence of PhSiD_3 (1.0 equiv) in EtOH at 60 °C for 1 h afforded the deuterated product **31A** in 75% yield as a single diastereoisomer. When this reaction was performed in deuterated ethanol ($\text{CD}_3\text{CD}_2\text{OD}$), product **31B** was isolated in 75% yield and deuterium atoms were distributed between C3 (29% D) and C1 (17% D) (D/H $\approx$ 1/3 in total). This indicated that the radical-based reduction could occur by abstractions of protons from both C1 and C3 carbons via [1,4]- and [1,5]-HAT processes. On the other hand, when ketal **31C**, in which the protons at C1 and C3 position are less activated compared with ketone **31**, was used as the substrate, **31E-iso** bearing an opposite stereochemistry at C9 was obtained in 70% yield, together with its isomer **31D** in 9% yield. These results support our proposed intramolecular HAT reaction.

The total synthesis of (+)-waihonsene was achieved by using Lee's protocol to install the C3 methyl group. Treatment of ketone **32** with LiHMDS followed by reaction with MeI gave **33** in 90% yield. Further reaction of **33** with the methylene triphenylphosphonium ylide in refluxing toluene afforded product **6** in 90% yield on a 100 mg scale. The synthetic material was fully characterized, and its ^1H and ^{13}C NMR spectra and specific rotation were found to be identical to those of the natural product.²¹

■ TOTAL SYNTHESIS OF (–)-SPIROCHENSILIDE A (7)

In 2015, Gao and co-workers isolated spirochensilide A³⁴ (**7**, Figure 6) from *Abies chensiensis*, which is an endemic Chinese plant. Biological investigation shows that compound **7** inhibits NO production, with 30% inhibition at a concentration of 12.5 $\mu\text{g}/\text{mL}$, suggesting that **7** might be a useful probe for studying inflammatory diseases.³⁵

Spirochensilide A (**7**) represents a member of an emerging and biologically important class of natural products, featuring an unusual spiro[4.5]ring system (BC ring), an anomeric spiroketal (EF ring), and two pairs of vicinal all-carbon quaternary chiral centers (C8/C10 and C13/17). The highly rigid scaffold of **7** renders it a challenging target for total synthesis. In 2020, we reported an asymmetric total synthesis of **7**,² and the key steps in

the synthesis are a semipinacol rearrangement and a tungsten-mediated cyclopropane-based PK reaction.

On the basis of a retrosynthetic analysis, we proposed an oxidative cyclization³⁶ to build the EF spiroketal ring of **7** from furan **C-1**, which was expected to be obtained via an Aldol reaction of aldehyde **C-2** with enone **C-3**, followed by stereoselective installations of the two vicinal methyl groups at C17 and C18. The unusual BC spiro[4.5]ring system in **C-3** was expected to be constructed via a PK reaction of enyne **C-4** and a semipinacol rearrangement³⁷ of epoxide **C-6** as key steps. The challenging AB bicyclic ring system in **C-7** could potentially be built up via a biomimetic cyclization³⁸ of the functionalized isoprenoid polyene **C-8**.

Scheme 4 shows our total synthesis of **7**. Our first innovation involved the development of a biomimetic cyclization of epoxide **34** for enantioselective generation of bicycle **35** bearing a quaternary carbon at C10 and a vinyl bromide. The biomimetic cyclization of **34** was achieved by treatment with TiCl_4 (0.4 equiv) in CH_2Br_2 at -35°C for 1 h; vinyl bromide **35** was obtained in 90% yield. Further coupling of the vinyl bromide moiety in **35** with acetylene via a Sonogashira reaction, and subsequent silylation with TBSCl furnished **36** in 88% overall yield.

Compound **36** was first subjected to a diastereoselective epoxidation by treatment with *m*CPBA, the resultant epoxide then underwent a semipinacol rearrangement to give **38** in 65% yield. After alkylation of **38** with lithium reagent **39** in the presence of CeCl_3 at -98°C , the resultant secondary alcohol was protected with TES to afford **40** in 74% overall yield. However, PK reaction of enyne **40** could not afford the desired product **41b** under the conventional PK reaction [with $\text{Co}_2(\text{CO})_8$] and in PK-type reactions [with other metal complexes derived from Rh, Pd, Ir, or Ru]. We later found that use of $\text{W}(\text{CO})_3(\text{MeCN})_3$ as the catalyst enabled isolation of **41b** in ca. 30% yield, together with its diastereoisomer **41a** in 30% yield. We also tested other catalysts, such as $\text{Ni}(\text{COD})_2/\text{bipy}$ or $\text{Mo}(\text{CO})_3(\text{DMF})_3$; all these reactions provided **41a** and **41b** but favored **41a**; the overall yields were higher.

Regioselective cyclopropane opening was achieved by treating **41b** with $^t\text{BuOK}$ to selectively remove its TMS group, and the resultant product **42** first proceeded by a Pd/C-catalyzed regioselective hydrogenation to remove the double bond at C14–C15, followed by a reductive ring-opening reaction of the resultant product **43** with Li/NH_3 to give **44** in 76% yield over three steps after quenching with DCE.

To install the *trans*-double bond at C17–C20 in enone **47**, ketone **44** was reacted with $^t\text{BuBOTf}/\text{DIPEA}$, and the newly formed silyl enol-ether then participated in a highly diastereoselective enol-borane-mediated aldol reaction with TBS-stabilized furyl acetaldehyde **45** to give hydroxyketone **46**, which was then reacted with 2-fluoro-1-methylpyridin-1-ium tosylate followed by neutral Al_2O_3 -mediated *syn*-elimination to give enone **47** in 73% overall yield.

We then turned our attention to achieving the diastereoselective synthesis of the allylic alcohol motif in **50**. To this end, enone **47** first underwent a diastereoselective cuprate-mediated 1,4-addition by treatment with Me_2CuLi , and the resultant ketone was methylated by reaction with MeI in the presence with KH to afford ketone **48** bearing the desired C17 and C20 stereogenic centers. Ketone **48** was then treated with lithium diisopropylamide (LDA) followed by reaction with PhSeCl to give a selenide **49**, which finally underwent a selective oxidation

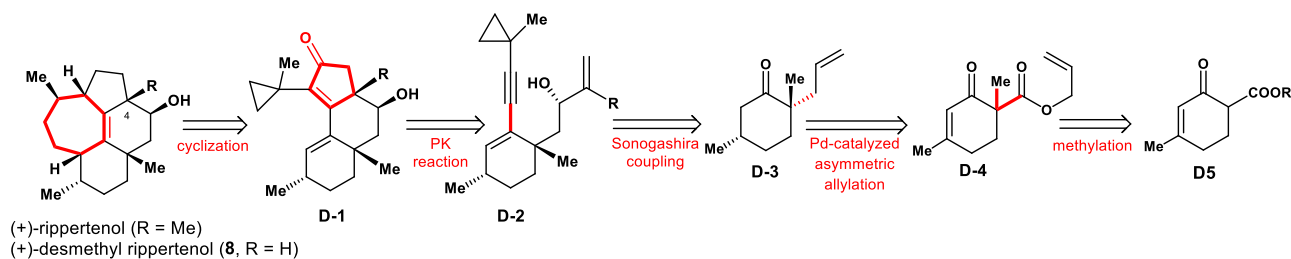
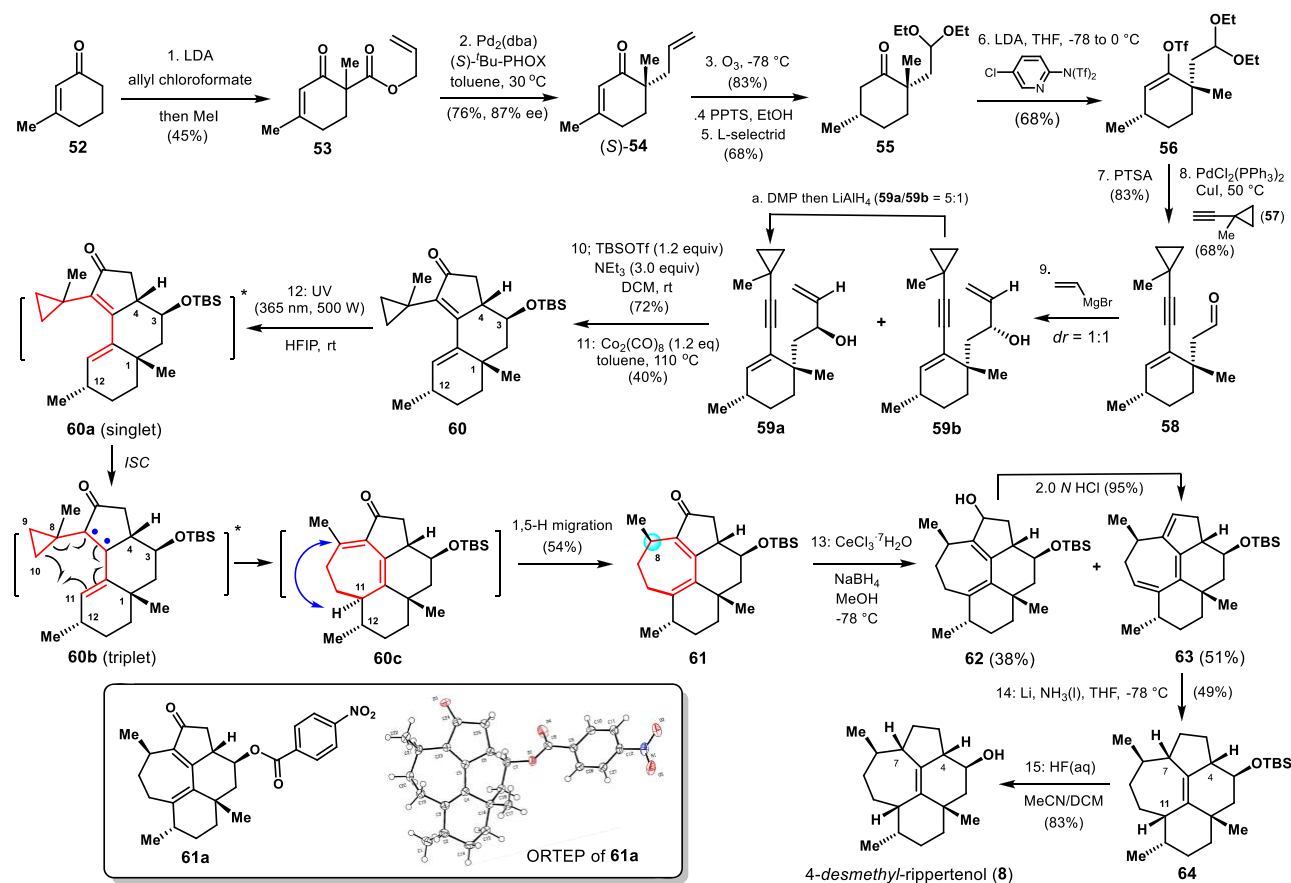


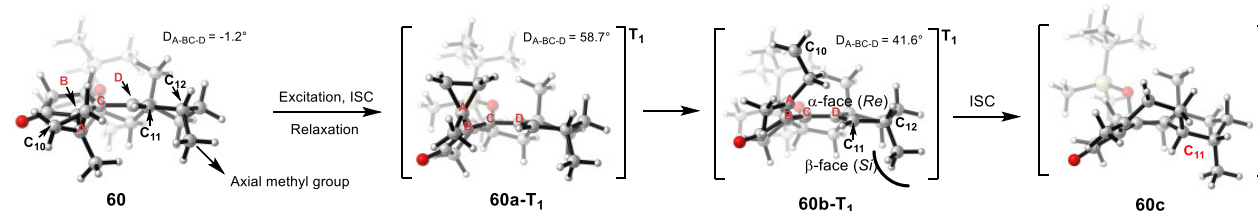
Figure 7. Retrosynthetic analysis of (+)-4-desmethyl-rippertenol (8).

Scheme 5. Total Synthesis of (+)-4-Desmethyl-rippertenol (8) and DFT-Evaluation of the Photoinduced Cyclization of 60

A: Total synthesis of (+)-4-desmethyl-rippertenol (8)



B: Evaluation of the photo-induced cyclization



and reduction sequence to afford **50** in 66% yield over three steps.

We next moved toward total synthesis. Upon treatment of **50** with singlet oxygen (generated by irradiation of oxygen with a tungsten lamp in the presence of methylene blue), the furan motif of **50** was oxidized to 4-oxo-2-alkenoic acid, which was treated with ClCH₂CO₂H in MeCN to afford **51** in 88% yield. Selective desilylation of **51** with TBAF·3H₂O and subsequent

oxidation of the newly generated secondary alcohol with Dess–Martin periodinane (DMP) followed by a desilylation with HF gave the target molecule **7** in 87% yield over three steps. The structure of synthetic (–)-spirochensilide A was confirmed by its NMR spectroscopy, optical rotation data, and single-crystal X-ray diffraction. Our first round of the synthesis produced more than 150 mg of **7**.

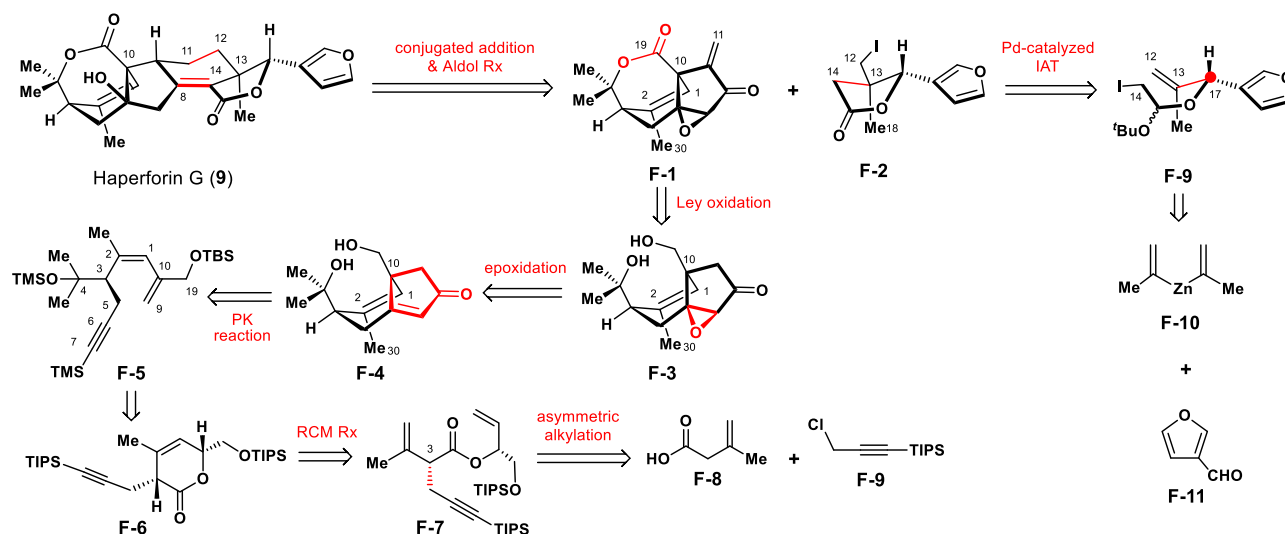


Figure 8. Retrosynthetic analysis of (+)-haperforin G (9).

TOTAL SYNTHESIS OF (+)-4-DESMETHYL-RIPPERTENOL

Rippertenol (Figure 7)³⁹ was isolated from the defensive secretions of the higher termites *Nasutitermes rippertii* and *Nasutitermes ephratae*, and this compound features a unique [6,6,5,7] bowl scaffold with one tetrasubstituted double bond and seven stereogenic centers, two of which are quaternary carbons. Rippertenol shows potent activity against methicillin-resistant *Staphylococcus aureus*. From a synthetic point of view, the absence of functionality makes its stereoselective total synthesis more challenging.

In 2009, the Metz group completed an asymmetric synthesis of 4-desmethyl-rippertenol (8)⁴⁰ through reaction sequences involving a ring expansion of cyclohexanone to cycloheptanone and an intramolecular Diels–Alder reaction. In 2011, the Snyder group presented the first total synthesis of rippertenol by using an inverse electron-demand Diels–Alder reaction and late-stage ring expansion to forge the fused cycloheptanoid structure.⁴¹ In 2020, we reported our effort to develop a concise approach to the asymmetric total synthesis of 4-desmethyl-rippertenol (8) with PK reaction and UV-light-induced cyclization reaction as the key steps.³

Figure 7 shows our retrosynthetic analysis. We envisioned that the cyclohepta-1,3-diene motif in 8 could be constructed via a UV-light-induced cyclization of α -cyclopropyl-substituted dienone D-1. The core in D-1 was expected to be derived from enyne D-2 via an intramolecular PK reaction. The conjugated enyne in D-2 could be obtained from ketone D-3, which could be prepared from ketoester D-4 through a Pd-catalyzed asymmetric allylation. Our retrosynthetic analysis therefore traced back to enone D-5, which can be obtained via a well-established protocol.

Our synthesis started with an enantioselective preparation of enyne 58a (Scheme 5). Starting from commercially available 3-methyl-2-cyclohexen-1-one 52, successive nucleophilic substitutions with allyl chloroformate and MeI in the presence of a base gave β -ketoester 53. A Pd-catalyzed enantioconvergent decarboxylative allylation developed by Stoltz et al.⁴² gave olefin (S)-54 in 76% yield with 87% ee.

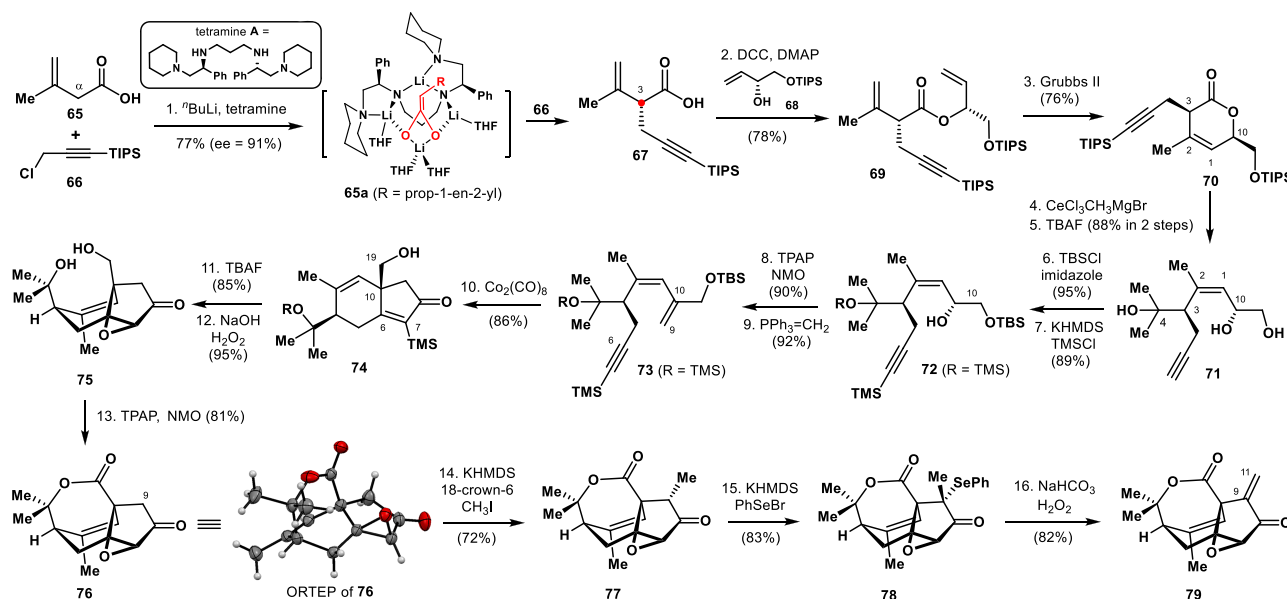
After ozonization of 54, the resultant aldehyde was converted to its corresponding acetal, and 1,4-reduction with bulky L-selectride gave cyclohexanone 55 as a single diastereoisomer.

Treatment of 55 with Comins' reagent gave vinyl trifluoromethanesulfonate 56 in 68% yield. The acetal motif was removed by acid-mediated hydrolysis, and the resultant aldehyde was subjected to a Sonogashira coupling reaction with 1-ethynyl-1-methyl-cyclopropane 57 to afford aldehyde 58. Nucleophilic addition of the aldehyde with vinyl magnesium bromide afforded an inseparable mixture of diastereoisomers 59a and 59b in a 1:1 ratio. Subjecting enyne 59a to Co-promoted PK reaction gave dienone 60 bearing a [6,6,5] tricyclic ring in 40% yield as a single diastereoisomer. The structure of 60 was confirmed by X-ray crystallographic analysis of its derivative. The unwanted isomer 59b was converted to 59a by sequential oxidation and reduction to give a 5:1 mixture of 59a/59b, that is, favoring the desired product 59a. With dienone 60 in hand, we tested the proposed photoinduced cyclization. When 60 was irradiated with UV light (365 nm, 500 W) in the presence of hexafluoroisopropanol (HFIP) at room temperature, 61 was obtained in 54% yield as a single diastereoisomer. This could be formed through the proposed photoinduced cyclization to generate 60c and a subsequent unexpected 1,5-hydrogen migration through a concerted suprafacial process.⁴³ The structure of 61 was confirmed by X-ray crystallographic analysis of its derivative 61a.

Computational experiments were performed to clarify the mechanism of the stereocontrolled photoinduced conversion of 60 to 60c (Scheme 5B). Because of occupied orbital repulsion of the triplet excited state 60a-T₁ (before relaxation), excitation and subsequent intersystem crossing (ISC) of 60 led to a significant conformational change (after relaxation) in the dienone moiety; the dihedral angle D_{A-BC-D} changed to 58.7° from -1.2°. Cyclopropane ring-opening affords diradical 60b-T₁ (D_{A-BC-D} = 41.6°), which cyclizes to 60c via a second ISC. These results indicate that the stereoselectivity of this photoinduced process can be mainly attributed to the relaxed conformation of the excited dienone moiety, in which the C10–C11 bond forms from the α -face, which has less torsion strain. Repulsion between C10 and the axial methyl group on C12 prevents C10 from approaching C11 from the β -face. These two factors enable stereospecific formation of the C11 stereocenter.

With 61 in hand, we explored the total synthesis of 8. Luche reduction of 61 resulted in the formation of allylic alcohol 62 and triene 63 in 38% and 51% yields, respectively. Alcohol 62

Scheme 6. Synthesis of Lactone 78



was converted to **63** under acidic conditions (2.0 N HCl) in 95% yield. Birch reduction of **63** afforded **64** in 49% yield. Finally, **8** was obtained in 83% yield by the treatment of **64** with HF in a mixed MeCN and CH₂Cl₂ solvent. The spectroscopic and optical rotation data of the synthesized **8** matched those reported by Hennig and Metz.⁴⁰

■ TOTAL SYNTHESIS OF (+)-HAPERFORIN G

(+)-Haperforin G (**9**, Figure 8), a flagship member of this biologically important class of limonoid tetranortriterpenoid natural product family, was isolated as a trace compound (37 mg/25 kg) from *Harrisonia perforate*.⁴⁴ Since its initial isolation in 2001, **9** has been demonstrated to inhibit human 11 β -hydroxysteroid dehydrogenase type 1 (11 β -HSD1) (IC₅₀ 0.58 μ M),⁴⁵ suggesting its potential as an agent for treatment of Alzheimer's disease, vascular inflammation, cardiovascular disease, and glaucoma.⁴⁶ Structurally, **9** is characterized by a novel limonoid 6/5/6 tricyclic carbon framework featuring six stereogenic centers, two of which are all-carbon quaternary stereogenic centers, and two lactones and a 3-substituted furan ring.

Figure 8 shows our retrosynthetic analysis. We envisaged that **9** could be generated sequentially via a convergent coupling of enone **F-1** with iodide **F-2** to form the C11–C12 bond, followed by a sequence of an intramolecular aldol reaction and a dehydration reaction to assemble its polycyclic core. We envisioned that the seven-membered lactone ring of **F-1** could be installed via oxidation of diol **F-3** by taking advantage of its favorable boat-like conformation. The epoxide motif in diol **F-3** could be derived via an epoxidation from **F4**, which in turn could be obtained from enyne **F-5** by an intramolecular PK reaction. This PK reaction could be accelerated by the TMS-protected tertiary alcohol moiety (C3) due to its Thorpe–Ingold effect. The enyne **F-5** could be readily accessed from lactone **F-6** via a double cerium(III)-mediated Grignard reaction. We also envisaged that the *cis*-double bond (C1–C2) in lactone **F-6** could be achieved regioselectively from diene **F-7** via the ring-closing metathesis (RCM) reaction developed by Roulland and Ermolenko.⁴⁷ Thus, our retrosynthetic analysis would trace back to synthesis of diene **F-7**, which was expected to be achieved via

Zakarian's highly enantioselective alkylation⁴⁸ of the commercially available vinyl acetic acid **F-8** with chloride **F-9**.

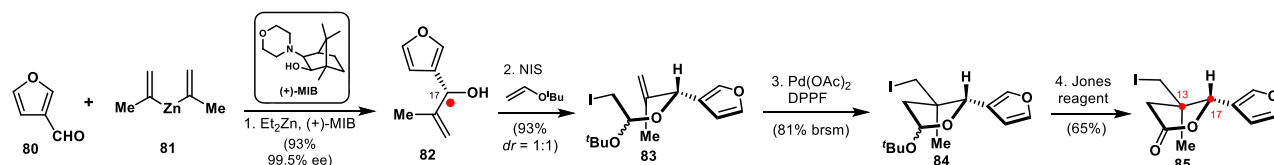
For diastereoselective installation of the C13 all-carbon quaternary chiral center in **F-2**, we would like to adopt a Pd-catalyzed iodide-atom transfer cyclization⁴⁹ (IATC) of alkenyl iodide **F-9**. **F-9** was expected to be made from 3-exo-morpholinoisoborneol (MIB)-mediated enantioselective nucleophilic addition of the vinyl-zinc agent **F-10** to aldehyde **F-11**.⁵⁰ Use of these state-of-the-art synthetic methodologies minimized synthetic manipulations ancillary to construction of the molecular framework of haperforin G, which ensured overall synthetic efficiency.

Our research began with the stereoselective synthesis of triol **71** (Scheme 6). Acid **65** was treated with ^tBuLi (4.0 equiv) in the presence of chiral tetramine **A** (1.05 equiv) at –78 °C, and the resultant complex **65a** was reacted with chloride **66** (1.2 equiv) producing acid **67**, which bears an *R*-configured C3 stereogenic center, in 77% yield and with 91% ee, together with 13% of the γ -alkylation isomer. To prepare diene **69**, acid **67** was coupled with alcohol **68** in the presence of dicyclohexylcarbodiimide (DCC)/4-dimethylaminopyridine (DMAP), and the formed diene **69** then underwent a Grubbs-II-catalyzed diastereomer-discriminating RCM reaction⁵¹ to afford lactone **70** as a single diastereoisomer in 59% overall yield. Reaction of **70** with excess methyl magnesium bromide in the presence of CeCl₃ and then desilylation with TBAF gave triol **71** in 88% yield.

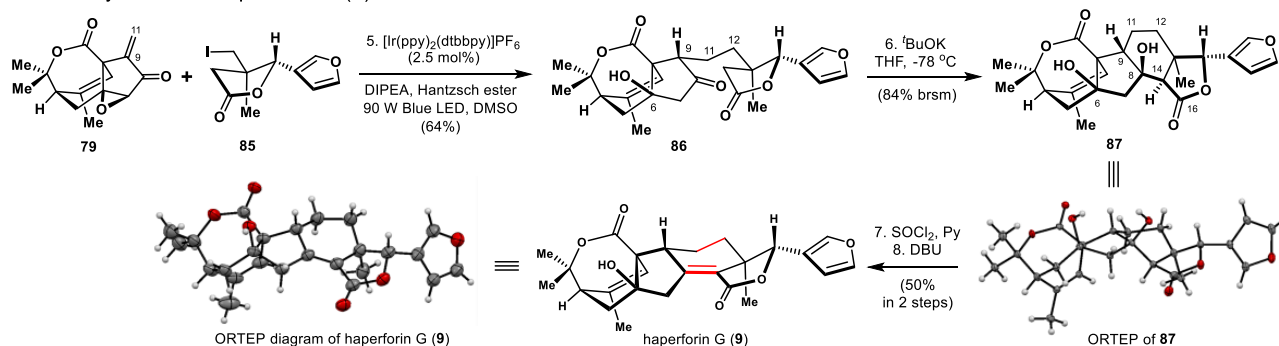
We then tested our proposed PK reaction as a key step for diastereoselective synthesis of enone **76**. With this aim, **71** was first converted to **72** in 85% yield by sequential silylation and desilylation. Oxidation of the secondary alcohol in **72** with tetrapropylammonium perruthenate (TPAP) and *N*-methylmorpholine *N*-oxide (NMO) produced the corresponding ketone, which underwent a Wittig reaction to afford **73** in 83% yield. Further treatment of enyne **73** in the presence of a catalytic amount of Co₂(CO)₈ (20 mol %) under a balloon pressure of CO at 110 °C for 36 h gave enone **74** in 86% yield. Thus, enone **74** was treated with TBAF, and the resultant diol was subjected to nucleophilic epoxidation by treatment with H₂O₂ in the presence of NaOH to give epoxide **75** (81% overall

Scheme 7. Synthesis of Iodide 85 and Total Synthesis of (+)-Haperforin G (9)

A: Synthesis of iodide 85.



B: Total synthesis of haperforin G (9)



yield) in a regio- and diastereoselective manner. Finally, use of Ley's oxidation resulted in direct conversion of 75 to the corresponding lactone 76 in 81% yield.

To prepare enone 79, ketone 76 was treated with potassium bis(trimethylsilyl)amide (KHMDs) in the presence of 18-crown-6, and the resultant enolate was reacted with MeI to give 77 in 72% yield, together with its *O*-alkylated isomer in 17% yield. Further treatment of 77 with KHMDs afforded an enolate, which was then reacted with PhSeBr to give 78 in 83% yield. Thus, upon oxidation of 78 with H₂O₂ in the presence of NaHCO₃, enone 79 was obtained in 82% yield.

We next turned our attention to the synthesis of iodolactone 85 (Scheme 7A). Aldehyde 80 was reacted with vinylzinc 81 under Nugent's (+)-MIB-mediated enantioselective vinylation conditions to give allylic alcohol 82 in 93% yield, with excellent ee. The *N*-iodosuccinimide (NIS)-mediated intermolecular ketalization of 82 with *tert*-butyl vinyl ether produced iodide 83 in 93% yield. Thus, under optimization of the reaction conditions, iodide 83 was converted to iodide 84 via a Pd-catalyzed IATC reaction in 81% yield. Finally, a Jones' oxidation of 84 led to formation of iodolactone 85 in 65% yield.

With compounds 79 and 85 in hand, we entered the final stage of the total synthesis as shown in Scheme 7B. Inspired by recently reported photoredox-catalyzed C–C bond formations,^{52–54} we intended to couple enone 79 with iodide 85 via this photochemistry. To this end, in the presence of a catalytic amount of transition-metal chromophore, that is, [Ir(ppy)₂(dtbbpy)]PF₆* (2.5 mol %), under blue light⁵⁵ and using a Hantzsch ester as a reducing agent, the unactivated C(sp³)-based⁵⁶ iodide 85 was converted to the corresponding alkyl radical, which was intermolecularly coupled with enone 79 to produce ketoester 86 as a single diastereoisomer in 64% yield. Further treatment of 86 with *t*-BuOK in THF at –78 °C effected an intramolecular aldol condensation to give 87 as a single stereoisomer in 84% yield. Finally, taking advantage of the presence of a hydroxyl group at C-6, reaction of 87 with SOCl₂/pyridine produced a cyclic sulfite, which was then treated with 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) to give 9 in 50% overall yield. The ¹H and ¹³C NMR spectra and specific rotation of the synthesized 9 were in agreement with those reported in

the literature,⁴ and its structure has been confirmed by X-ray crystallographic analysis. The asymmetric total synthesis of 9 was therefore achieved in 20 steps, which included a concise Co-catalyzed PK reaction and a convergent photoredox catalysis as key steps.⁴

■ CONCLUSION

Natural products, which have an extensive track record of contributions to society, still have untapped potential in the field of chemistry, chemical biology, and drug discovery because of their interesting biological activities and natural scarcity. The establishment of efficient methods for synthesizing natural products not only enables access to synthetic analogs with improved potency and pharmacological activities but can also give rise to tool compounds, such as molecular probes for mechanistic studies. In the past two decades, to accelerate progress, our group has systematically investigated PK reactions for generating the scaffolds of architecturally complex structures. Tailoring of these frameworks by using our well-defined PK reactions hastened to the assembly of challenging targets, for example, (–)-crinipellin A (5), (+)-waihoensene (6), (–)-spirochensilide A (7), (+)-4-desmethyl-rippertenol (8), and (+)-haperforin G (9). Implementation of the PK reaction, on the basis of our understanding of the inherent topology and innate reactivity of PK reaction precursors or by the judicious use of theoretical studies has extended the repertoire of complex and diastereoselective organic syntheses. The wide utility and enormous potential of the PK reaction is deemed worthy of pursuit because its successful implementation will productively extend its already expansive frontiers and prove valuable as we strive to mimic nature in the efficient synthesis of organic molecules.

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Notes

The author declares no competing financial interest.

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